THE RESOLUTION PROPERTY FOR ONE DEFINING SHEAF OF IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 13573-13709, immediately following the statement at 13565-13571.

Editorial note. Long proof: the approximation stage cites fourteen cross-chapter pointers (Limits of Spaces, Descent on Spaces, Derived Categories of Spaces); all of them are navigation pointers recorded in assembly.external_references with their canonical tags.

Editorial note. No step of the author's deduction is outside the span; the Noetherian case is invoked only after the approximation has been constructed inside the span.

Allowed standard prerequisites (editorial).

- The situation of descending the resolution property. chapter situation situation-descend-resolution-property, cited in the statement and used throughout. Fixes the setup: a quasi-compact quasi-separated algebraic space with a finite locally free covering by an affine scheme and the classifying ideal I.
- Noetherian case of the resolution property. chapter lemma lemma-resolution-property-in-Noetherian-case, cited at proof line 13703. Supplies the conclusion after the descent argument has produced a Noetherian approximation with the same data.
- Pushforward of a finite module along a finite morphism. Descent on Spaces, lemma spaces-descent-lemma-finite-over-finite-module, cited at proof lines 13577-13578. Keeps finite type quasi-coherent modules finite type under the pushforward along the finite locally free covering.
- Approximation of an algebraic space by a Noetherian one. Limits of Spaces, proposition spaces-limits-proposition-approximate with the descent lemmas listed in the card, cited from proof line 13690 onward. Realises the space, the covering and the relevant module over a Noetherian base so that the Noetherian case applies.
- Base change of the resolution property along affine morphisms. Derived Categories of Spaces, lemma spaces-perfect-lemma-resolution-property-goes-up-affine, cited at proof lines 13707-13708. Transfers the property from the Noetherian approximation back to X.

Source reference. Original lemma 0GV0: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13538–13538. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GUY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13416–13416. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted

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proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07U7: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03I5: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 060Z: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0850: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SL: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084Z: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SK: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0D2X: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 11–11. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07V7: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 12–12. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07VR: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 13–13. Navigation pointer only: the counted

proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SQ: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 14–14. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 07SU: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 15–15. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GUS: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 16–16. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GUU: focused-seeds/geometry/S9-spaces-resolve-prop-one-sheaf/ref-tags.txt, source lines 17–17. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 0GUY, the algebraic space X has the resolution property if and only if $\pi_*\mathcal{I}$ is the quotient of a finite locally free $\mathcal{O}_X$-module.*

Proof. The pushforward $\pi_*\mathcal{G}$ of a finite type quasi-coherent $\mathcal{O}_Y$-module $\mathcal{G}$ is a finite type quasi-coherent $\mathcal{O}_X$-module by Descent on Spaces, Lemma 060Z. In particular, if X has the resolution property, then $\pi_*\mathcal{I}$ is the quotient of a finite locally free $\mathcal{O}_X$-module by Derived Categories of Spaces, Definition 0GUS.

Assume that we have a surjection $\mathcal{E} \to \pi_*\mathcal{I}$ for some finite locally free $\mathcal{O}_X$-module $\mathcal{E}$. In the rest of the proof we show that X has the resolution property by reducing to the Noetherian case handled in Lemma 0GV0. We suggest the reader skip the rest of the proof.

A first reduction is that we may view X as an algebraic space over $\mathrm{Spec}(\mathbf{Z})$, see Spaces, Definition 03I5. (This doesn't affect the conditions nor the conclusion of the lemma.)

By Limits of Spaces, Lemma 07VR we can write $Y = \lim Y_i$ with Y_i finite and of finite presentation over X and where the transition maps are closed immersions. Consider the closed subspace $Z = V(\mathcal{I})$ of Y. Since $\mathcal{I}$ is of finite type, the morphism $Z \to Y$ is of finite presentation. Hence we can find an i and a morphism $Z_i \to Y_i$ of finite presentation whose base change to Y is $Z \to Y$, see Limits of Spaces, Lemma 07SK. For $i' \geq i$ denote $Z_{i'} = Z_i \times_{Y_i} Y_{i'}$. After increasing i we may assume $Z_i \to Y_i$ is a closed immersion (of finite presentation), see Limits of Spaces, Lemma 0850. Denote $\mathcal{I}_i \subset \mathcal{O}_{Y_i}$ the ideal sheaf of Z_i and denote $\pi_i : Y_i \to X$ the structure morphism. Similarly for $i' \geq i$. Since $Z = \lim_{i' \geq i} Z_{i'}$ we have

$$\pi_*\mathcal{I} = \mathrm{colim}\, \pi_{i',*}\mathcal{I}_{i'}$$

The transition maps in the system are all surjective as follows from the surjectivity of the maps $\pi_{i,*}\mathcal{O}_{Y_i} \to \pi_{i',*}\mathcal{O}_{Y_{i'}}$ and the fact that $Z_{i'} = Z_i \times_{Y_i} Y_{i'}$. By Cohomology

of Spaces, Lemma 07U7 for some $i' \geq i$ the map $\mathcal{E} \to \pi_*\mathcal{I}$ lifts to a map $\mathcal{E} \to \pi_{i',*}\mathcal{I}_{i'}$. After increasing i' this map $\mathcal{E} \to \pi_{i',*}\mathcal{I}_{i'}$ becomes surjective (since if not the colimit of the cokernels, having surjective transition maps, is nonzero). This reduces us to the case discussed in the next paragraph.

Assume X is an algebraic space over $\mathbf{Z}$ and that $Y \to X$ is of finite presentation. By absolute Noetherian approximation we can write $X = \lim X_i$ as a directed limit, where each X_i is a quasi-separated algebraic space of finite type over $\mathbf{Z}$ and the transition morphisms are affine, see Limits of Spaces, Proposition 07SU. Since $\pi : Y \to X$ is of finite presentation we can find an i and a morphism $\pi_i : Y_i \to X_i$ of finite presentation whose base change to X is π, see Limits of Spaces, Lemma 07SK. After increasing i we may assume π_i is finite, see Limits of Spaces, Lemma 084Z. Next, we may assume there exists a finite locally free $\mathcal{O}_{X_i}$-module $\mathcal{E}_i$ whose pullback to X is $\mathcal{E}$, see Limits of Spaces, Lemma 0D2X. We may also assume there is a map $\mathcal{E}_i \to \pi_{i,*}\mathcal{O}_{Y_i}$ whose pullback to X is the composition $\mathcal{E} \to \pi_*\mathcal{I} \to \pi_*\mathcal{O}_Y$, see Limits of Spaces, Lemma 07V7. The cokernel

$$\mathcal{E}_i \to \pi_{i,*}\mathcal{O}_{Y_i} \to \mathcal{Q}_i \to 0$$

is a coherent $\mathcal{O}_{Y_i}$-module whose pullback to X is the (finitely presented) cokernel $\mathcal{Q}$ of the map $\mathcal{E} \to \pi_*\mathcal{O}_Y$. In other words, we have $\mathcal{Q} = \pi_*(\mathcal{O}_Y/\mathcal{I})$. Consider the map

$$\mathcal{E}_i \otimes_{\mathcal{O}_{X_i}} \pi_{i,*}\mathcal{O}_{Y_i} \longrightarrow \pi_{i,*}\mathcal{O}_{Y_i} \otimes_{\mathcal{O}_{X_i}} \pi_{i,*}\mathcal{O}_{Y_i} \to \pi_{i,*}\mathcal{O}_{Y_i} \to \mathcal{Q}_i$$

where the second arrow is given by the algebra structure on $\pi_{i,*}\mathcal{O}_{Y_i}$. The pullback of this map to Y is zero because the image of $\mathcal{E} \to \pi_*\mathcal{O}_Y$ is the ideal $\pi_*\mathcal{I}$. Hence by Limits of Spaces, Lemma 07V7 after increasing i we may assume the displayed composition is zero. This exactly means that the imag of $\mathcal{E}_i \to \pi_{i,*}\mathcal{O}_{Y_i}$ is of the form $\pi_{i,*}\mathcal{I}_i$ for some coherent ideal sheaf $\mathcal{I}_i \subset \mathcal{O}_{Y_i}$. Since $\mathcal{E}_i \to \pi_{i,*}\mathcal{O}_{Y_i}$ pulls back to $\mathcal{E} \to \pi_*\mathcal{O}_Y$ we see that the pullback of $\mathcal{I}_i$ to Y generates $\mathcal{I}$. Denote $V_i \subset Y_i$ the open subspace whose complement is $V(\mathcal{I}_i) \subset Y_i$. Then V is the inverse image of V_i by the comments above. After increasing i we may assume that V_i is affine and that $\pi_i|_{V_i} : V_i \to X_i$ is étale, see Limits of Spaces, Lemmas 07SQ and 07SL. Having said all of this, we may apply Lemma 0GV0 to conclude that X_i has the resolution property. Since $X \to X_i$ is affine we conclude that X has the resolution property too by Derived Categories of Spaces, Lemma 0GUU. □